

Status

RK3588 / Orange Pi 5 Max — A/B-virt Emulator — Status

Revision: 2 **Last modified:** 2026-06-11T12:30:00Z **Scope:** The hardware-free emulation ladder that exercises the Helix OTA native Android A/B update flow (update_engine + AVB/dm-verity + auto-rollback) for RK3588 / Orange Pi 5 Max targets, on a developer host with NO live board. **Authority:** §11.4.45 (integration Status doc), §11.4.5 (captured-evidence table), §11.4.6 (no-guessing — pending work is stated as PENDING, never implied done), §11.4.107 (liveness, not a single frame), §11.4.112 (structurally-gated tiers SKIP honestly). **HEAD at last update:** 0d27438.

Operator-blocked / Pending items (read first — §11.4.45(9))

Item	State	What is needed	§-ref
A/B slot switch + dm-verity + auto-rollback (the real OTA-apply flow)	PENDING_FORENSICS — assembler + boot.cmd AUTHORED, not yet run	NOT yet proven. The base guest boots; the A/B disk assembler (assemble_ab_disk.sh) + U-Boot bootcount slot-select script (uboot_ab/boot.cmd) are now AUTHORED (parse-clean + internally coherent) but have NOT been run — they are gated on a real u-boot.bin, which the in-flight U-Boot+RAUC image build (build8) will produce. NO captured slot-switch / rollback evidence yet.	§11.4.6
Tier-2 Cuttlefish — REAL	SKIP on this host	No /dev/kvm on this Apple-Silicon macOS dev host (confirmed	§11.4.3 / §11.4.112

Item	State	What is needed	§-ref
Android A/B OTA		absent). Ready to RUN on the operator's incoming Linux + nested-KVM host. NOT a fake PASS.	
Tier-3 — real RK3588 / Orange Pi 5 Max hardware	PENDING — no board	Requires physical hardware on the bench.	§11.4.6

Fidelity ladder (honest)

The emulation tiers trade fidelity for hardware-freedom. Higher tiers exercise more of the real OTA stack but demand scarcer environments.

Tier	What it proves	Fidelity	State
T0 — Containerized control-plane + device emulator (podman)	Control-plane $\leftrightarrow$ device round-trip (register $\rightarrow$ update-check $\rightarrow$ telemetry) with NO live hardware	Protocol / control-plane only — no real A/B apply	SHIPPED
T1 — A/B-virt base image on QEMU virt + HVF	The aarch64 guest boots to a live interactive Linux userspace on the Apple CPU — the foundation the A/B layers build on	Real kernel + userspace boot; A/B-apply layers AUTHORED (assembler + U-Boot boot.cmd), NOT yet run	FOUNDATION GREEN (boot); A/B layers AUTHORED, pending u-boot.bin (build8 in flight)
T2 — Cuttlefish (cvd) on Linux + nested KVM	The real Android update_engine A/B + AVB/dm-verity + auto-rollback apply flow	Closest hardware-free proxy for the RK3588 OTA apply	SKIP on this host (no /dev/kvm); ready for Linux host
T3 — Real RK3588 / Orange Pi 5 Max board	The genuine on-device OTA apply + AVB/dm-verity + bootloader rollback	Full fidelity	PENDING — no hardware

Captured-evidence status table (§11.4.5)

Closed verdict vocabulary: PASS / FAIL / SKIP / PENDING_FORENSICS / OPERATOR-BLOCKED. Every PASS/SKIP cites a real evidence path verified to exist in this repo at revision time.

#	Capability under test	Verdict	Evidence (real, repo-relative)	No
E1	PWU-AB-0/1 base guest image built (aarch64 Buildroot, kernel 6.1.44 + ext2 rootfs)	PASS	tests/emulator/ab_virt/out/images/Image (MD5 9f3670cd7dba7bdeeebe2c6d791e929a), tests/emulator/ab_virt/out/images/rootfs.ext2 (MD5 a056760e88eea575977be13e38cfe430); builder tests/emulator/ab_virt/build_image.sh	The con PW arti Bo reb wit -de hos fro 274 flig rep ado boo rov arti at r
E2	PWU-AB-1 FOUNDATION — boots to LIVE interactive userspace on QEMU virt + HVF	PASS	docs/qa/20260611T061626Z-ab-virt-boot/console.log (196 lines) — kernel boots on Apple CPU (physical CPU 0x0000000000 [0x610f0000], MIDR 0x610f), buildroot login: root (line 174), post-login sentinel HELIX_USERSPACE_LIVE_OK (line 182, emitted only from an interactive shell after login), clean poweroff.	§11 live boo pos sen sin Dri emu ab boo
E3	A/B slot switch (slot A→B select)	PENDING_FORENSICS	Tooling AUTHORED (not run): A/B disk assembler tests/emulator/ab_virt/assemble_ab_disk.sh + U-Boot bootcount slot-select tests/emulator/ab_virt/uboot_ab/boot.cmd (+ uboot.env, README.md) — both parse-clean + coherent. NO captured slot-switch evidence yet.	NO Ga u-l (bu flig per §11 fak
E4	dm-verity integrity on the booted slot	PENDING_FORENSICS	— (RAUC dm-verity slot layers in build8; not yet exercised)	De No
E5	Auto-rollback on failed boot (bootcount → fall back to	PENDING_FORENSICS	Logic AUTHORED in uboot_ab/boot.cmd (bootcount / bootlimit / altbootcmd / BOOT_ORDER semantics per U-Boot Boot Count Limit) — NOT run	De E4 exe

#	Capability under test	Verdict	Evidence (real, repo-relative)	No
E6	known-good slot)	PASS	Driver tests/emulator/tier1_container_e2e.sh (boots ota-server + ota-device-emu in a podman pod; asserts register → update-check → telemetry from captured container logs under docs/qa/<run-id>/)	Pro pla onl NC rea
	T0 containerized control-plane ⇔ device round-trip			§11 §11 top (ex fak scr ma OT rol inv UN per Lin run the inc hos
E7	T2 Cuttlefish — REAL Android A/B OTA apply (incl. PWU-CF-2 corrupt-slot auto-rollback)	SKIP — UNVERIFIED pending Linux host	Driver tests/emulator/tier2_cuttlefish_ab.sh (PWU-CF-2 headline corrupt-slot → reboot → auto-rollback section authored, mirrors ab_virt PWU-AB-3); host gate: /dev/kvm absent on this Apple-Silicon macOS host (verified)	No ber
E8	T3 real RK3588 / Orange Pi 5 Max hardware	PENDING_FORENSICS	—	Sta arti run any wh evi mis NC exe app tha evi is v bar
E9	HelixQA bank — helix_ota.yaml static gate (incl. HOTA-AB-VIRT-B00T)	PASS	bash tools/helixqa/run_bank.sh --dry-run --bank tools/helixqa/banks/helix_ota.yaml → 12 passed / 0 failed / 0 skipped ; HOTA-AB-VIRT-B00T declares evidence docs/qa/20260611T061626Z-ab-virt-boot/console.log (the same E2 boot transcript, verified present)	

What is genuinely proven today (§11.4.6)

- **Proven (captured evidence):** the A/B-virt base guest image **builds** and **boots to a live, interactive Linux userspace** on QEMU virt + HVF on this Apple-Silicon host (E1, E2) — the foundation tier.
- **AUTHORED but NOT run (honestly pending):** the real A/B slot switch, dm-verity, and auto-rollback flow (E3–E5). The A/B disk assembler (`assemble_ab_disk.sh`) and the U-Boot bootcount slot-select script (`uboot_ab/boot.cmd`) are written, parse-clean, and internally coherent, but have NOT been executed — they are gated on a real `u-boot.bin`, which the in-flight U-Boot+RAUC image build (build8) will produce. There is no slot-switch or rollback evidence yet, so nothing here is marked proven.
- **Bank static gate (proven):** the HelixQA `helix_ota.yaml` bank passes its `--dry-run` evidence-artifact audit 12/0/0 (E9), with HOTA-AB-VIRT-B00T wired to the verified E2 boot transcript.
- **Honestly skipped (topology-gated):** the Tier-2 Cuttlefish real-Android-A/B path including the PWU-CF-2 corrupt-slot auto-rollback section (E7) SKIPS on this host for lack of `/dev/kvm` and is UNVERIFIED-pending a Linux + nested-KVM host. No fabricated continuity, no fake PASS.

Provenance

Commit	Subject
dd43738	research + dev-host A/B-virt build infra (base image build in progress)
d5374d0	PWU-AB-1 foundation GREEN — base image boots to LIVE userspace on QEMU+HVF
4278aa9	add U-Boot <code>qemu_arm64</code> + RAUC + GPT tooling to the A/B-virt build (PWU-AB-1 full, in progress)
0f120a6	parallel streams — HelixQA bank wiring (HOTA-AB-VIRT-B00T) + these §11.4.45 Status docs + Cuttlefish corrupt-slot rollback (PWU-CF-2)
6581ab4	A/B disk assembler (<code>assemble_ab_disk.sh</code>) + U-Boot bootcount slot-select (<code>uboot_ab/boot.cmd</code>) — PWU-AB-1, AUTHORED
0d27438	U-Boot host tools need <code>libssl-dev</code> (+bison/flex) — build7 failed on missing <code>openssl/evp.h</code> ; build8 in flight